Lab 4

Second-Order Systems, and Unity Feedback, PD and PV Controllers

Responsible Lab TA: Adam Bigsby

Group 06

Team members:
1. Saud Amjad (UCID: 30170966)
2. Hamzah Nadeem (UCID: 30180744)
3. Temuulen Ankhbayar (UCID: 30188632)

Lab dates: Oct 16, 19, 23, 26
Report due date: Oct 29, 2026

Team Contribution Table:

Team member	Experimental work and data collection	Analysis	MATLAB/ Simulink Code	Interpretation and Validation	Report Writing and Editing	Overall Contribution
Member 1 (Saud Amjad):	**S.A.**	**S.A.**	**S.A.**	**S.A.**	**S.A.**	33.33%
Member 2 (Hamzah Nadeem):	**H.N**	**H.N**	**H.N**	**H.N**	**H.N**	33.33%
Member 3 (Temuulen Ankhbayar):	**T.A**	**T.A**	**T.A**	**T.A**	**T.A**	33.34%
TOTAL:						100%

Acknowledgement: By submitting this report, all team members confirm that the information above accurately represents their contributions and that all team members have reviewed and approved the final report and associated files

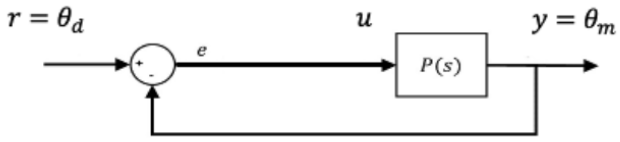
6.2: Second Order System Step Response with the Unity Feedback:

1. Derivation:

From the lab manual provided, the standard second-order transfer function has the form:

$$T(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

For the Unity Feedback Loop, we can use the closed-loop transfer function derived and equate it to the equation above to solve for ω_n and ζ:



The closed-loop transfer function:

$$\frac{\theta_m}{\theta_d} = \frac{(\frac{K}{\tau})}{s^2+(\frac{1}{\tau})s+(\frac{K}{\tau})} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

Where, $K = 23.1\ rad/s/V$ and $\tau = 0.165\ s$

We can then solve for the natural frequency (ω_n) and damping ratio (ζ):

$$\omega_n^2 = \frac{K}{\tau} \rightarrow \omega_n = \sqrt{\frac{K}{\tau}}$$

$$\omega_n = \sqrt{\frac{(23.1\ rad/s/V)}{0.165\ s}}$$

$$\boxed{\omega_n \approx 11.832 \frac{rad}{s}}$$

$$2\zeta\omega_n = \frac{1}{\tau} \rightarrow \zeta = \frac{1}{2\times\omega_n\times\tau}$$

$$\zeta = \frac{1}{2\times(11.832\frac{rad}{s}\times 0.165\ s)}$$

$$\boxed{\zeta \approx 0.256}$$

Natural Frequency (ω_n): $11.832\ \frac{rad}{s}$

Damping Ratio (ζ): 0.256

2. Peak Time and Percent Overshoot:

From the Lab Manual, the Peak Time and Percent Overshoot formulas with respect to the determined values in 6.2.1 (ω_n and ζ) have been provided and can be calculated:

Peak Time:

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{(11.832)\sqrt{1-(0.256)^2}}$$

$$t_p = 0.27\ s$$

Peak Overshoot (P.O.):

$$P.O. = 100e^{\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)} = 100e^{\left(-\frac{\pi(0.256)}{\sqrt{1-(0.256)^2}}\right)}$$

$$P.O. = 43.5\ \%$$

Expected Peak Time (t_p): $0.27\ s$

Percent Overshoot (P.O.): $43.5\ \%$

5.

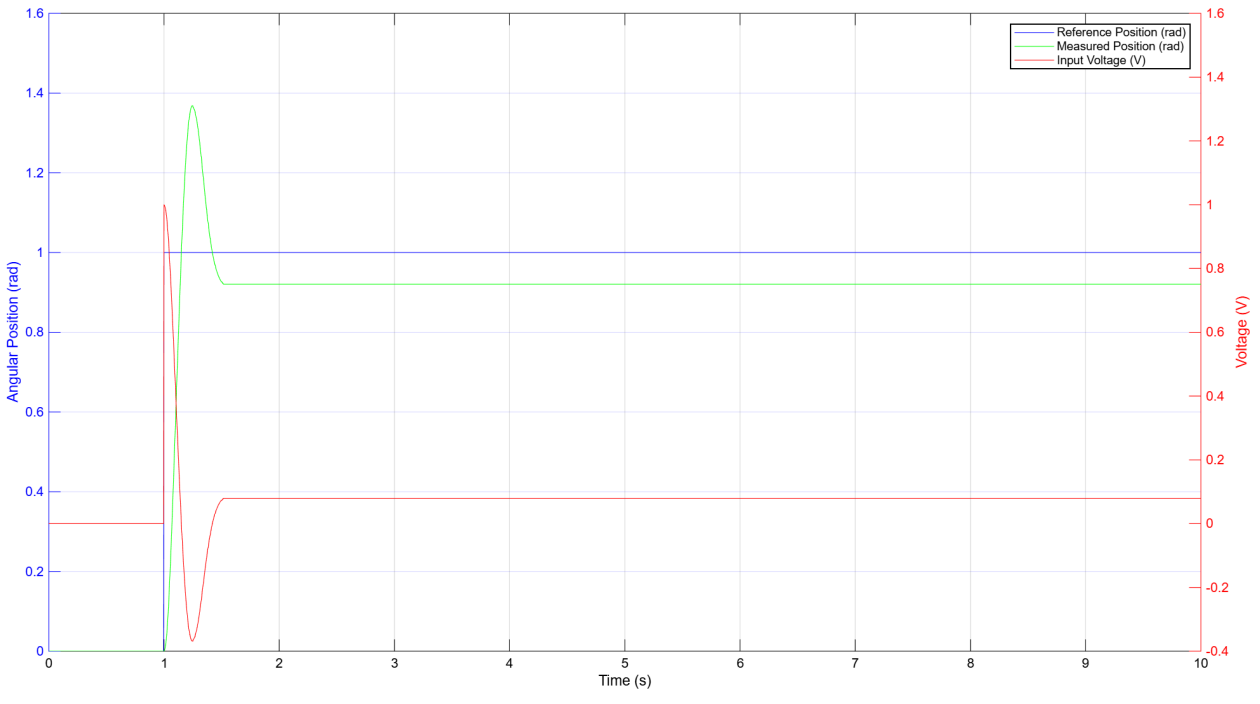


Figure 01: Unity Feedback Control - Step Response

6.

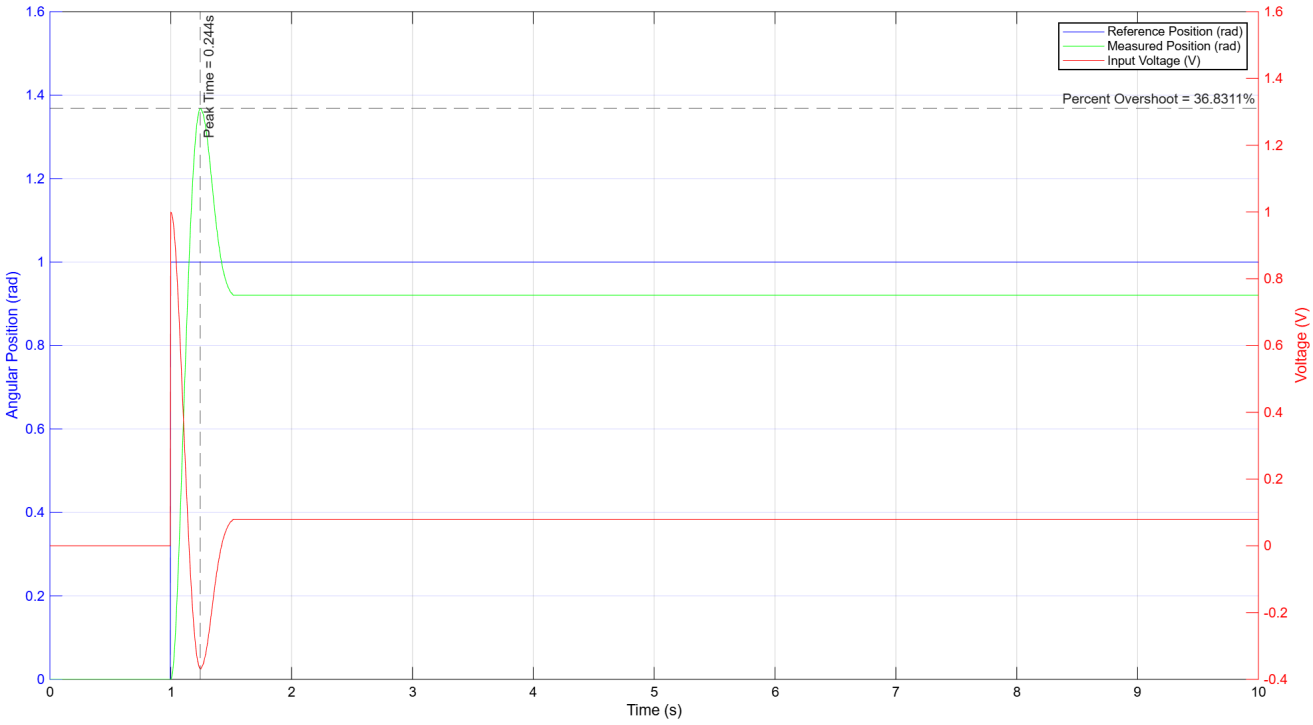


Figure 02: Unity Feedback Control - Set Point And Measured Position, with Markings for PO and t_p

The t_p is 0.244s compared with the predicted 0.27s giving us a difference of 0.026s (9.6% error). The Percent Overshoot (P.O) was experimentally determined to be 36.83% compared to the higher predicted P.O of 43.5% giving us a difference of 6.67% (15.3% error with respect to the theoretical P.O). The steady state of the measured graph is ~0.92 rad instead of the 1 rad which is due to the Coulomb friction of the motor. The voltage required to move the motor ~0.08 rad (from the graph, this is ~0.09V) is less than the voltage required to break the inherent static friction of the motor, which from Lab 3, was determined to be 0.19V. This means the motor cannot move to try to reach 1 rad.

7.1: Design Requirements and Constraints:

Table 1: Design Requirements and Constraints

Item	Our Entry	Source or Justification
Percent overshoot, P.O	$\leq 2.5\ \%$	Given
Peak time, t_p	$\leq 0.15\ s$	Given
Peak motor voltage, $\lvert V_m \rvert$	Must not saturate	Given
Amplifier saturation limit (V)	+/-10 V Without the Stall Detection Block +/-5 V With the Stall Detection Block	QUBE-Servo 2 User Manual Table 2.2 +/-15 V is the absolute maximum but must stay within the recommended +/-10 V. However, the Stall detection Block limits the voltage limit to +/-5V
Reference signal: amplitude and frequency	Trials 1-7: Square wave [-1, 1] rad at 0.4 Hz Trials 8-15: Square wave [-0.5, 0.5] rad at 0.4 Hz	We arbitrarily set the reference using the lab manual, and then updated it to reduce the risk of reaching the voltage stall limit
Encoder resolution, and the smallest angle you can resolve	2048 lines/rev. 0.00307 rad/count	QUBE-Servo 2 User Manual Table 2.2
Filter cut-off frequency, ω_f	100 rad/s	Lab 4 Manual (Section 6.3): Indicated that the Quanser reference implementation of 100 rad/s. Our group decided to continue with this value after conducting 6.3.
K and τ computed from your Lab 3 parameters	K = 24.41346647 τ = 0.120095272	Lab 3 Report Section 6.3.8 We converted the Speed over voltage transfer function into position over voltage.
ω_n and ζ implied by the specification above	ζ = 0.761323316 $\omega_n = 32.30234668\ rad/s$	Calculated using P.O and t_p
k_p and k_d implied by that ω_n and ζ	k_p = 5.132921334477 k_d = 0.20099124761191	Lab 4 Report Section 7.2.1 Calculated using the ζ, ω_n, τ, and K

7.2: Predict:

1. Derivation:

We can then use the desired P.O (2.5%) provided and determine the corresponding damping ratio:

$$P.O = 100e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\zeta = \pm\frac{|ln(\frac{PO}{100})|}{\sqrt{ln^2(\frac{PO}{100})+\pi^2}} = \pm\frac{|ln(\frac{2.5}{100})|}{\sqrt{ln^2(\frac{2.5}{100})+\pi^2}} = 0.761323316$$

Take the positive damping ratio:

$$\zeta = 0.761$$

Using the calculated damping ratio and the desired peak time, t_p, to solve for the ω_n.

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$\omega_n = \frac{\pi}{t_p\sqrt{1-\zeta^2}} = \frac{\pi}{(0.15)\sqrt{1-(-0.761323316)^2}} = 32.30234668\ rad/s$$

$$\omega_n = 32.302\ rad/s$$

From 6.2.1, the following values for K and τ were provided:

$$K = 23.1\ rad/s/V$$
$$\tau = 0.165\ s$$

From the lab manual provided, the standard second-order transfer function has the form:

$$T(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

For the PV feedback control the following closed loop expression for $Y(s)/R(s)$ is provided in the lab manual:

$$\frac{Y(s)}{R(s)} = \frac{Kk_p}{\tau s^2+(1+Kk_d)s+Kk_p} = \frac{(\frac{Kk_p}{\tau})}{s^2+(\frac{1+Kk_d}{\tau})s+(\frac{Kk_p}{\tau})}$$

Equating the two expressions, we can isolate for k_p and k_d:

$$\omega_n^2 = \frac{Kk_p}{\tau} \rightarrow k_p = \frac{\omega_n^2 \tau}{K}$$

$$2\zeta\omega_n = \frac{1+Kk_d}{\tau} \rightarrow k_d = (2\zeta\omega_n \tau - 1)\frac{1}{K}$$

We can solve for k_p and k_d:

$$k_p = \frac{\omega_n^2 \tau}{K} = \frac{(32.30234668)^2(0.165)}{(23.1)} = 7.45315429311$$

$$\boxed{k_p = 7.453}$$

$$k_d = (2\zeta\omega_n \tau - 1)\frac{1}{K} = (2(0.761323316)(0.165)(32.30234668) - 1)\frac{1}{(23.1)} = 0.30803180941$$

$$\boxed{k_d = 0.308}$$

Summary of determined values (using Lab 3 values for k_m and R_m):

Target Natural Frequency [ω_n] (rad/s)	32.302
Damping Ratio (ζ)	0.761
k_p	7.453
k_d	0.308

2.

From Lab 3, we can utilize the experimentally determined values for k_m and R_m, in order to find K and τ.

$$\frac{\theta_m(s)}{V_m(s)} = \frac{\frac{1}{s}\Omega_m(s)}{V_m(s)} = \frac{K}{s(\tau s+1)}$$

$$\frac{\Omega_m(s)}{V_m(s)} = \frac{K}{\tau s+1} = \frac{\frac{1}{k_m}}{\frac{R_m J_{eq}}{k_t k_m}s+1} \Rightarrow K = \frac{1}{k_m},\ \tau = \frac{R_m J_{eq}}{k_t k_m}$$

$$K = \frac{1}{k_m} = \frac{1}{0.040961} = 24.41346647$$

$$\tau = \frac{R_m J_{eq}}{k_t k_m} = \frac{(9.8862)(2.089856\times 10^{-5})}{(0.042)(0.040961)} = 0.120095272$$

$$K = 24.413$$
$$\tau = 0.120$$

The values for ζ and ω_n were previously determined above using the P.O and t_p indicated in Table 1 (previously determined in 7.2.1):

$$\zeta = 0.761$$
$$\omega_n = 32.302\ rad/s$$

Just as before, the derivation for k_p and k_d is provided again below. From the lab manual provided, the standard second-order transfer function has the form:

$$T(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

For the PV feedback control the following closed loop expression for $Y(s)/R(s)$ is provided in the lab manual:

$$\frac{Y(s)}{R(s)} = \frac{Kk_p}{\tau s^2+(1+Kk_d)s+Kk_p} = \frac{(\frac{Kk_p}{\tau})}{s^2+(\frac{1+Kk_d}{\tau})s+(\frac{Kk_p}{\tau})}$$

Equating the two expressions, we can isolate for k_p and k_d:

$$\omega_n^2 = \frac{Kk_p}{\tau} \rightarrow k_p = \frac{\omega_n^2 \tau}{K}$$

$$2\zeta\omega_n = \frac{1+Kk_d}{\tau} \rightarrow k_d = (2\zeta\omega_n \tau - 1)\frac{1}{K}$$

We can solve for k_p and k_d:

$$k_p = \frac{\omega_n^2 \tau}{K} = \frac{(32.30234668)^2(0.120095272)}{(24.41346647)} = 5.13292133448$$

$$\boxed{k_p = 5.133}$$

$$k_d = (2\zeta\omega_n \tau - 1)\frac{1}{K} = (2(0.761323316)(0.120095272)(32.30234668) - 1)\frac{1}{(24.41346647)}$$

$$k_d = 0.200991247612$$

$$\boxed{k_d = 0.201}$$

Summary of determined values (using Lab 3 values for k_m and R_m):

K (rad/s/V)	24.413
τ (s)	0.120

Target Natural Frequency [ω_n] (rad/s)	32.302
Damping Ratio (ζ)	0.761
k_p	5.133
k_d	0.201

The experimental proportional gain k_p is 5.133 using the Lab 3 values compared to the value of 7.453 using the provided K and τ values given in 6.2.1, giving us a difference of 2.32 or

$k_{p(using\ Lab\ 3)}$ is 31% smaller than $k_{p(using\ given\ values\ in\ 6.2.1)}$. The experimental derivative gain, k_d is 0.201 using the Lab 3 values compared to the value of 0.308 using the provided K and τ values given in 6.2.1, giving us a difference of 0.107 or $k_{d(using\ Lab\ 3)}$ is 35% smaller than $k_{d(using\ given\ values\ in\ 6.2.1)}$. In our PV controller, we will use the experimental values calculated for k_d and k_p derived from the Lab 3 data collected (7.2.2). This is because the experimental values (calculated from by using our experimental R_m and k_m) provides a more accurate representation of our specific motor, which will help when tuning the PV controller.

7.3: Test and Revise:

For the following table outlining our test runs we can use the following formulae for Percent Overshoot (P.O) and Peak Time (t_p):

$$PO_{predicted} = 100e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$t_{p_{(predicted)}} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

Equating this expression to T(s), we can get the following equations to solve for ζ and ω_n:

$$\omega_n = \sqrt{\frac{Kk_p}{\tau}}$$

$$2\zeta\omega_n = \frac{1+Kk_d}{\tau} \rightarrow \zeta = \frac{1+Kk_d}{2\omega_n\tau}$$

Recall from 7.2.2,

K	24.413
τ	0.120

We can now determine our values for $PO_{predicted}$ and $t_{p_{(predicted)}}$ as we adjust our k_d and k_p values.

4. Table 2: Input Reference: 1 rad

#	k_p	k_d	ωf	Predicted PO (%)	Predicted t_p(s)	Measured PO (%)	Measured t_p(s)	Peak V_m (V)	What we changed, and why
1	5.133	0.201	100	2.5	0.15	N/A	N/A	> 5V (Cannot measure exact voltage due to stall conditio n)	Reduced k_p since voltage crossed stall condition of 5V, meaning the gain amplifies the input voltage too high
2	1.133	0.201	100	N/A, damping ratio is 1.62	N/A, damping ratio is 1.62	Overdam ped therefore no PO	Overdam-ped therefore no t_p	2.0993	The system is overdamped, so we decreased k_d to decrease the damping ratio to get it under 1.
3	1.133	0.101	100	0.0065	0.67	Overdam ped therefore no PO	Overdam-ped therefore no t_p	2.14	The system is overdamped, so we increased k_p to amplify the gain past the reference position
4	2.383	0.101	100	6.54	0.19	8.9	0.17	4.876	Decreased k_p to reduce percent overshoot and to reduce voltage applied to stay under 5V.
5	1.883	0.101	100	3.24	0.24	4	0.25	3.853	The system is slightly underdamped with a long t_p, so we will decrease the damping ratio by decreasing k_d
6	1.883	0.076	100	9.04	0.20	16	0.19	3.9165	The system is slightly underdamped with a larger P.O, so we will decrease k_p to decrease P.O.
7	1.883	0.12	100	0.83	0.29	2.5	0.30	3.83	The system is slightly underdamped with a longer t_p, so we want to increase k_p to lower t_p.

8	2.2	0.12	100	2.16	0.23	2.8	0.28	4.468	We decrease to a step of 1 (-0.5 to 0.5) as we want to increase k_p to lower t_p, but we will hit the stall limit of 5V if we exceed k_p of ~2.5 using our current input voltage square wave of a step of 2 (-1 to 1).
9	2.5	0.12	100	3.64	0.20	6.1	0.237	2.57	The system is underdamped with a longer t_p, so we want to increase k_p to lower t_p.
10	3	0.12	100	6.21	0.17	6.76	0.16	3.11	The system is underdamped with a longer t_p, so we want to increase k_p to lower t_p.
11	3.3	0.12	100	7.73	0.16	10.44	0.137	3.43	The system is underdamped with a higher P.O, so we want to increase k_d to lower P.O
12	3.3	0.18	100	0.42	0.24	0.6	0.244	3.33	The system is underdamped with a longer t_p, so we want to decrease k_d to lower t_p.
13	3.3	0.15	100	2.86	0.18	2.46	0.20	3.32	The system is underdamped with a longer t_p, so we want to decrease k_d to lower t_p.
14	3.3	0.13	100	5.83	0.16	6.2	0.154	3.38	The system is underdamped with a higher P.O, so we want to increase k_d to lower P.O
15	3.3	0.14	100	4.21	0.17	3.08	0.177	3.35	Final Controller.

5. Our final design does not meet every row of Table 1. Our percent overshoot and peak time are slightly higher than the maximum specification values allowed. With more iterations, we could have achieved the desired results, as there is still some room to increase k_p and adjust k_d when looking at the motor stall voltage requirement. One reason for being unable to meet the requirements was because of our first change to the PV controller. Because the initial proportional gain was so large it resulted in the motor stalling, we reduced it by ~80% to ensure we stayed under the stall voltage. However, this did not work out well for us, as reducing the gain by that much put us much farther away from having a critically damped system, which is

around where our specifications could be met. If instead we started with a smaller step down of k_p, it would have been possible to reach our final controller gains in less iteration numbers, giving us more time to test out other proportional and derivative gains.

7.4: Trade-offs:

6. One of our most difficult aspects of the lab was learning how to balance speed against overshoot. For example, looking at trial 11, we can see that the percent overshoot is much higher than 2.5%, but our peak time fits the requirement of less than 0.15 seconds. To fix the issue of the percent overshoot being higher, we reduced k_d. While this did achieve the desired percent overshoot in Trial 12 (being 0.6%), it increased our peak time to 0.244 seconds due to having a higher damping ratio. Another consideration our group had to consider was the effect of damping against noise amplification. It was observed when increasing the derivative action from Trial 14 to 15, where we increased k_d from 0.13 to 0.14. With more derivative action, we have increased the amplification of noise. However, by increasing k_d, we have also increased damping ratio, ζ. With increasing derivative action to achieve a lower P.O (such as from Trial 14 to 15), our experiment shows that we have increased damping ratio at the expense of amplifying noise. The filter cut-off frequency can introduce some phase lag into the system as well. If the frequency of the signal is close to the cutoff frequency, then the system will respond with a time delay, which occurs due to a phase difference between the input signal to the filter and the output. However, we did not record the data right before the filter, so seeing the effects of the cut-off frequency and phase lag in our system is not possible.

7.

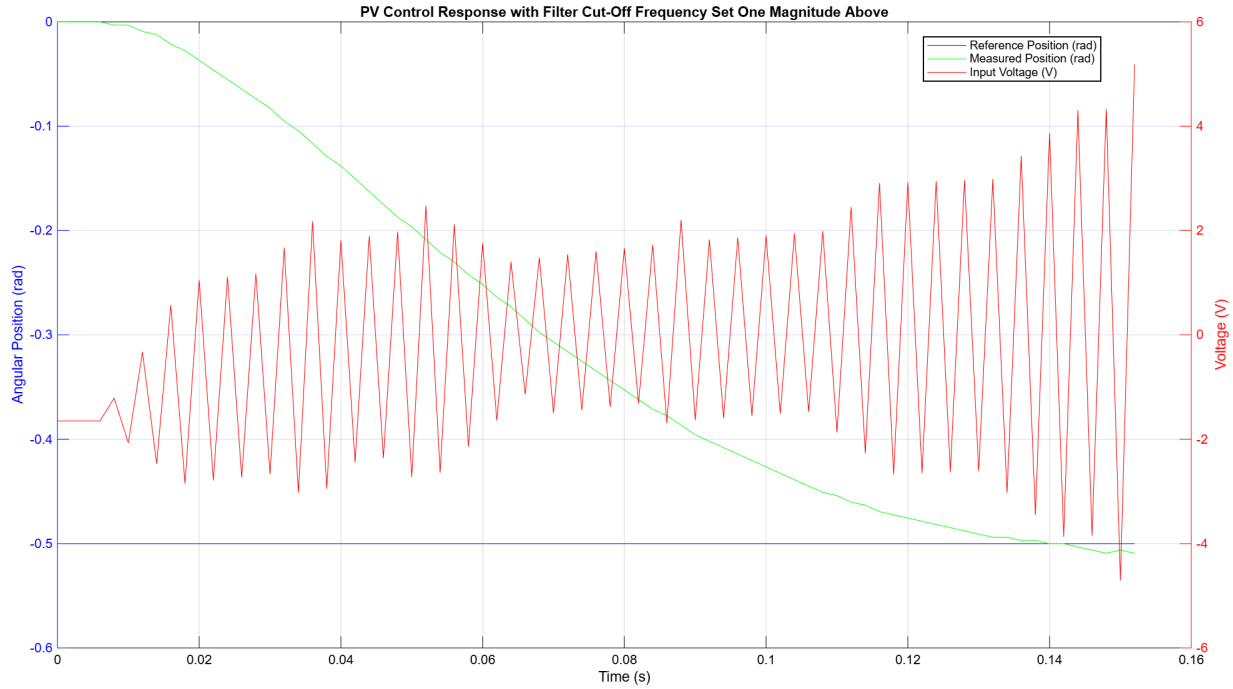


Figure 03: Filter Cut-off Frequency One Order of Magnitude Above

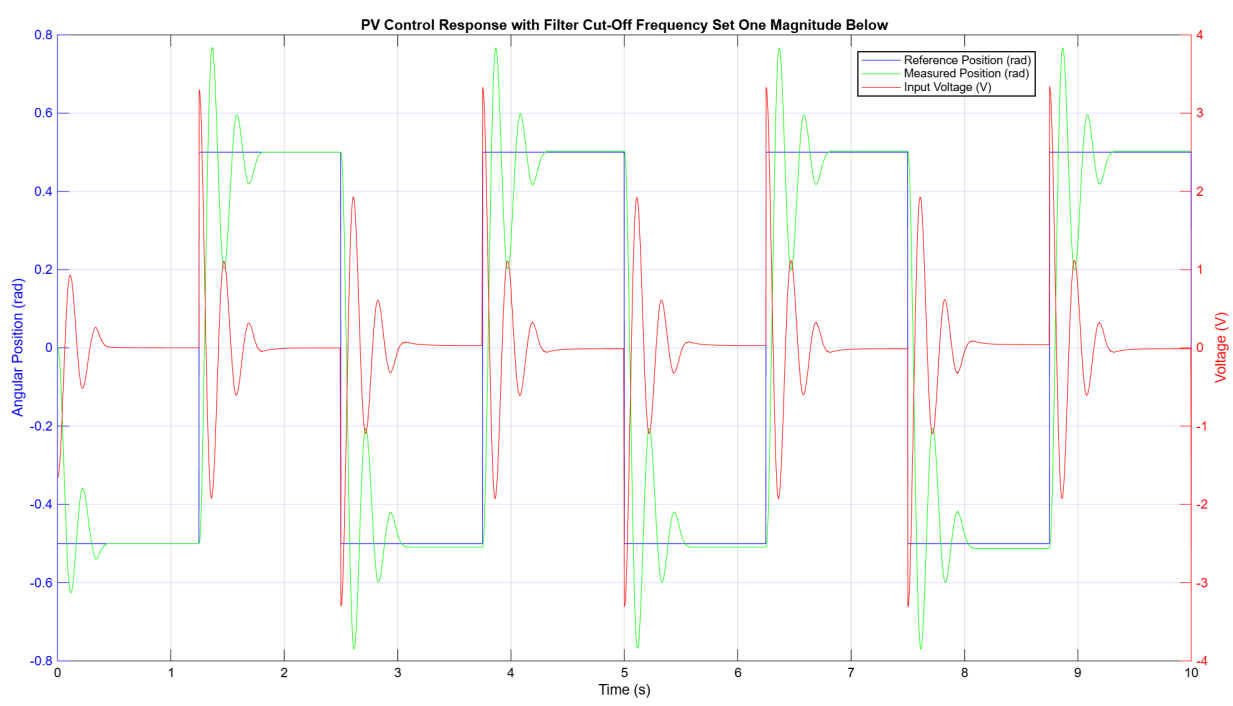


Figure 04: Filter Cut-off Frequency One Order of Magnitude Below

From the two plots above, we can see what the effects are from the noise. From the 'one magnitude above' plot, it can be seen that the noise from reading the position leaks into the input voltage calculation. The filter is set to 159.2 Hz, so more higher frequency noise below that cut-off frequency passes through the controller. We can see the impact of the encoder noise on the Angular Speed plot without the filter retrieved from Lab 2 [1] (Plot is below), where we see the amplification of noise after the derivative block (so the noise is much larger on the speed plot). We had utilized the filter to remove a lot of this noise. Because the filter is multiplied into the derivative action of the controller, the higher frequency noise gets amplified along with the signal. The system also becomes very overdamped, with the poles of the system shifting right into Quadrant 1. This is expected with a higher derivative action, as a higher k_d leads to a higher damping ratio. These oscillations make the system unstable, which leads to the input voltage going past the saturation limit of 5V of the stall detection block, as seen in the graph.

From the 'one magnitude below' plot, it can be seen that a phase lag has occurred. By decreasing the cutoff frequency, it is more likely that a phase shift will occur between the input signal and output. This can be seen in the graph with the input voltage and measured position. The first input voltage peak does not result in an immediate change in position. There is a time delay between the input voltage peak and the percent overshoot peak (around 0.12 seconds). At a filter frequency of one magnitude lower, the phase shift angle will be positive, and from the formula $\frac{\Delta t}{T} = \frac{\Delta \phi}{360°}$, this will mean that Δt is positive which indicates a time delay/lag. This was observed from the time delay when comparing the peaks of the input voltage and measured position plots.

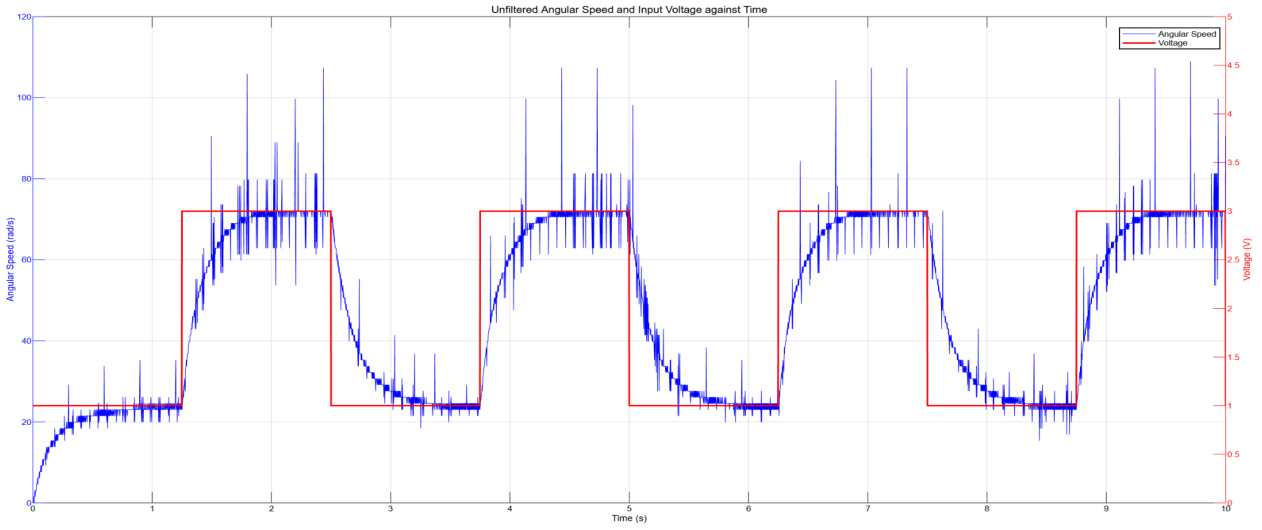


Figure 05: Unfiltered Angular Speed and Input Voltage against Time (retrieved from our Lab 02 report [1])

References:

[1] Saud Amjad. Hamzah Nadeem, Temuulen Ankhbayar. “Lab 2: Hardware Interfacing, Filtering, and Modelling” (2026)

- We used a plot from our Lab 2 report in 7.4.7 to indicate how encoder noise is amplified in our speed plot after the noise passed through a derivative block. We referenced this plot and showed it again above on Page 15.